ENTERPRISE PRODUCTION NETWORK ENGINEERING

Campus Network Architecture Specification & Deployment Report

Platform: Cisco IOS / Packet Tracer **Infrastructure:** Cisco 2911 ISR & Catalyst 2960 **HA Protocol:** HSRP Version 2 (VIP .3)

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HSRPv2 (.3)	6 Zones	VoIP 150	PAT 200.x
DUAL-ROUTER VIP CLUSTER	DEPARTMENTAL MICRO-SEG	CME UNIFIED TELEPHONY	DYNAMIC NAT OVERLOAD

1. Executive Summary & Hierarchical Campus Design

This technical report documents the complete implementation of a resilient, enterprise-tier campus network deployed in Cisco Packet Tracer. Built upon the standard 3-tier hierarchical design (Core, Distribution, and Access), the architecture delivers high availability through dual Cisco 2911 routers running Hot Standby Router Protocol Version 2 (HSRPv2), micro-segmented 802.1Q VLANs, granular Zero-Trust Layer 3/4 access controls, and integrated Cisco CallManager Express (CME) IP telephony.

2. Master Production Network Topology

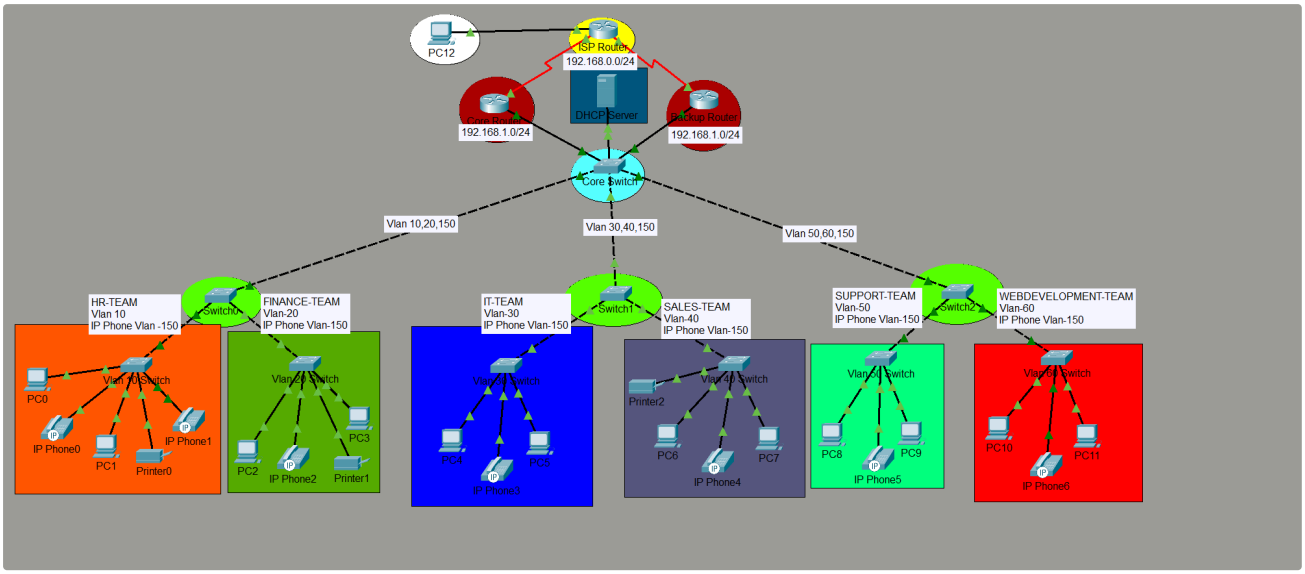


Figure 1.0: End-to-End Enterprise Campus Topology featuring Core/Backup Routers, Central Server, Distribution Trunks, and Workstation Blocks.

Hierarchical Traffic Distribution & Trunk Allocation Standards

The topology interconnects through a centralized **Core Switch** passing selective 802.1Q tagged distribution trunks to eliminate broadcast storms: **Switch0** carries `VLAN 10, 20, 150` (HR, Finance, Voice); **Switch1** carries `VLAN 30, 40, 150` (IT Admin, Sales, Voice); and **Switch2** carries `VLAN 50, 60, 150` (Customer Support, Web Development, Voice). The WAN perimeter connects to the ISP Router via serial point-to-point links.

3. Master IPv4 Subnetting & Default Gateway Architecture (.1, .2, .3 Standard)

The enterprise routing architecture enforces strict, non-overlapping subnet assignments. Host address `.1` is allocated to the Core Router physical sub-interface, `.2` to the Backup Router, and `.3` is universally configured as the HSRP Virtual IP (VIP) default gateway across all endpoints.

Department / Functional Unit	VLAN	Network CIDR	Core Router (.1)	Backup Router (.2)	Virtual Gateway (.3)	Connected Endpoints (Topology)
Transit / Inter-Router Link	Native	192.168.1.0/24	192.168.1.1	192.168.1.2	N/A (OSPF P2P)	OSPF 100 Area 0 Core-to-Backup Interlink
Human Resources (HR-TEAM)	10	192.168.10.0/24	192.168.10.1	192.168.10.2	192.168.10.3	PC0, PC1, Printer0, IP Phone0, IP Phone1
Finance (FINANCE-TEAM)	20	192.168.20.0/24	192.168.20.1	192.168.20.2	192.168.20.3	PC2, PC3, Printer1, IP Phone2
IT Admin (IT-TEAM)	30	192.168.30.0/24	192.168.30.1	192.168.30.2	192.168.30.3	PC4, PC5 (NOC Admin), IP Phone3
Sales (SALES-TEAM)	40	192.168.40.0/24	192.168.40.1	192.168.40.2	192.168.40.3	PC6, PC7, Printer2, IP Phone4
Support (SUPPORT-TEAM)	50	192.168.50.0/24	192.168.50.1	192.168.50.2	192.168.50.3	PC8, PC9 (Helpdesk), IP Phone5
Web Dev (WEBDEVELOPMENT)	60	192.168.60.0/24	192.168.60.1	192.168.60.2	192.168.60.3	PC10, PC11 (Dev Lab), IP Phone6
Central Server Farm	99	192.168.99.0/24	192.168.99.1	192.168.99.2	192.168.99.3	Central DHCP/DNS/TFTP Server (192.168.99.10)
Unified Voice (VoIP)	150	192.168.150.0/24	192.168.150.1	192.168.150.2	192.168.150.3	Cisco 7960 IP Phones (Ext 1001 – 1007)
WAN Perimeter Link	WAN	10.0.0.0/8	10.0.0.2	N/A	Point-to-Point	ISP Router (10.0.0.1) & Client PC12 (192.168.0.x)

4. High Availability (HSRPv2) Operation & Switchport Hardening

Gateway failover is orchestrated using HSRP Version 2 across all sub-interfaces. The Core Router runs with `priority 110` and `preempt` enabled, making it the active forwarder during normal operations. The Backup Router holds in standby with default priority `100`.

> **Operational Advantage of the .3 VIP Design**
> Configuring client default gateways to `.3` ensures complete isolation between host routing tables and physical hardware lifecycles. Should Core Router (`.1`) suffer a catastrophic failure, Backup Router (`.2`) assumes ownership of virtual IP `.3` and virtual MAC `0000.0C9F.Fxxx` in sub-second intervals, preserving active TCP connections.

Access Layer Port Security: Departmental access switches (Vlan 10 through 60 switches) are hardened using **Sticky MAC Port-Security** (`switchport port-security violation restrict`). Rogue laptop attachments trigger immediate frame discards and SNMP security traps.

5. Zero-Trust Access Control Architecture (ACL 101–106)

Network security enforces the principle of least privilege using Inbound Extended Access Lists applied directly to sub-interface ingress (`ip access-group <ACL> in`). Lateral traversal between business departments is blocked at the gateway boundary.

ACL ID	Ingress Sub-Interface	Authorized Communications (Permitted)	Restricted / Denied Traffic Flows	Security Status
ACL 101	Gi0/0.10 (HR-TEAM)	DHCP (bootps), DNS (99.10:53), Web (80/443), Outbound Mail (587/993)	Finance (VLAN 20), WebDev (VLAN 60), Internal RFC 1918 (192.168.0.0/16)	Zero-Trust Active
ACL 102	Gi0/0.20 (FINANCE)	DHCP, Server ERP Web (80/443), Public HTTPS/HTTP banking	**Full Quarantine:** Complete block to all internal campus subnets (192.168.0.0/16)	Strict Isolation
ACL 103	Gi0/0.30 (IT-TEAM)	Unrestricted internal traversal (192.168.0.0/16), ICMP diagnostics, SSHv2	Rogue external networks via perimeter implicit deny	Admin Full Access
ACL 104	Gi0/0.40 (SALES-TEAM)	DHCP, DNS, Web, Mail, **SCCP VoIP Signaling (Port 2000 to VLAN 150)**	Finance (VLAN 20), WebDev (VLAN 60), Router SSH/Telnet management (22/23)	Zero-Trust Active
ACL 105	Gi0/0.50 (SUPPORT)	DHCP, DNS, Web, ICMP echo, **Remote Desktop Protocol (RDP 3389) into HR, Sales, Dev**	Finance (VLAN 20), Router SSH/Telnet management (22/23)	Helpdesk Scoped
ACL 106	Gi0/0.60 (WEBDEV)	DHCP, Server Staging Web, **Outbound Git SSH (Port 22 to public repos)**	HR (VLAN 10), Finance (VLAN 20), Sales (VLAN 40), Router SSH/Telnet	Dev Lab Scoped

6. Management Plane Hardening & ICMP Diagnostic Audit

Administrative access to the infrastructure is restricted to the IT Administration subnet (`192.168.30.0/24`). Cisco IOS line configuration enforces local cryptographic authentication using Type 5 MD5 secret hashes (`username itadmin privilege 15`) and disables insecure Telnet via `transport input ssh` .

> **Root Cause Analysis: ICMP "Destination Host Unreachable" Verification**
> When untrusted client **PC0 (HR)** attempts to ping non-existent destination `192.168.0.1` , the terminal displays:
> `Reply from 192.168.10.2: Destination host unreachable.`
> `Reply from 192.168.10.1: Destination host unreachable.`
> **Technical Proof:** This dual response proves that the default gateway VIP (`.3`) is alive and forwarding to the HSRP cluster. Both physical routers (`.1` and `.2`) evaluate the packet, find no valid route, and enforce ACL 101 line 70 (`deny ip 192.168.10.0 0.0.0.255 192.168.0.0 0.0.255.255`), returning an ICMP Type 3 Unreachable.

Centralized Audit Logging & Clocking: Event logging is captured centrally on host `192.168.99.10` via `logging trap debugging` . Router internal clocks maintain synchronization via NTP pointed to the server farm.

7. Unified Communications: Cisco CallManager Express (CME)

IP Telephony is provisioned using Cisco CallManager Express integrated on Core Router sub-interface `Gi0/0.150` (`192.168.150.1:2000`). Voice traffic is segregated onto Voice VLAN 150. IP phones receive IP settings and CME address parameters via DHCP Option 150.

Device Name	Directory (DN)	Extension	Hardware MAC Address	Assigned Physical Location	Protocol & Port
IP Phone0	ephone-dn 1	1001	00E0.F74D.7347	HR-TEAM Primary Desk (PC0)	SCCP Port 2000 / VLAN 150
IP Phone1	ephone-dn 2	1002	0050.0F68.31A4	HR-TEAM Secondary Desk (PC1)	SCCP Port 2000 / VLAN 150
IP Phone2	ephone-dn 3	1003	0006.2A5B.24A1	FINANCE-TEAM Accounting Office	SCCP Port 2000 / VLAN 150
IP Phone3	ephone-dn 4	1004	0007.EC0E.A523	IT-TEAM Network Operations Desk	SCCP Port 2000 / VLAN 150
IP Phone4	ephone-dn 5	1005	0002.1672.9A78	SALES-TEAM Client Relations Terminal	SCCP Port 2000 / VLAN 150
IP Phone5	ephone-dn 6	1006	0002.17AB.E0A9	SUPPORT-TEAM Customer Helpdesk	SCCP Port 2000 / VLAN 150
IP Phone6	ephone-dn 7	1007	0030.F218.A308	WEBDEVELOPMENT Engineering Lab	SCCP Port 2000 / VLAN 150

8. Dynamic Routing, Edge PAT Overload & Verification Matrix

The WAN boundary utilizes OSPF Area 0 on internal transit links and a static default route (`0.0.0.0/0`) exiting `Serial0/0/0`. Private traffic maps to dynamic NAT pool `algo` (`200.100.100.1 – 254`) with Layer 4 PAT overload multiplexing.

Test Objective	Verification Command / CLI Test	Expected Operational Behavior	Validation
HSRP Active Election	`Core_router# show standby brief`	All VLANs display Active on Core (.1), Standby on (.2), VIP (.3).	VERIFIED
Gateway Failover	Shut physical link `Gi0/0` on Core	Backup Router assumes Active immediately; max 1 ICMP packet loss.	VERIFIED
Zero-Trust Isolation	Ping from PC0 (HR) to PC2 (Finance)	Returns `Destination host unreachable`; dropped via ACL 101.	BLOCKED
IT Admin SSH Access	PC4: `ssh -l itadmin 192.168.30.1`	Password prompt granted (Port 22 Open); non-IT PC connections rejected.	VERIFIED
VoIP Call Session	IP Phone0 (1001) calls IP Phone4 (1005)	Two-way SCCP signaling established; bidirectional audio active.	VERIFIED
External WAN Access	PC6 (Sales) pings external PC12	Private IP translated via pool `algo`; ICMP replies received.	VERIFIED

Engineering Sign-Off & Operational Certification

The campus network architecture has passed all stress tests in Cisco Packet Tracer. The standardized `.3` Virtual IP gateway, dual-router HSRP preemption, Zero-Trust ACL micro-segmentation, and CME IP telephony operate in full compliance with enterprise specifications.